2021秋高数上期末试题.(回忆版)

1. Multiple Choice Questions: (only one correct answer for each of the following questions.)

(1) Let $f(x)$ be a continuous function on $[-a,a]$, $a>0$, then $\int_{-a}^{a} f(x)\,dx =$

(A) $\int_0^a (f(x)+f(-x))\,dx$.

(B) $\int_0^a (f(x)-f(-x))\,dx$.

(C) 0

(D) $2\int_0^a f(x)\,dx$.

(2) If $f(x)=\begin{cases}\frac{2+e^{\frac{1}{x}}}{1+e^{\frac{1}{x}}}, & x\neq 0\\ 0, & x=0\end{cases}$, then at $x=0$, it is a

(A) jump discontinuity.

(B) removable discontinuity

(C) infinite discontinuity.

(D) continuous point.

(3) If the function $f(x)$ has the third derivative at $x=x_0$, and $f'(x_0)=f''(x_0)=0$, $f^{(3)}(x_0)>0$, then

(A) $f(x)$ has a local minimum at x_0.

(B) $f(x)$ has a local maximum at x_0.

(C) $f(x)$ has no local extremum at x_0.

(D) None of (A), (B) and (C) is correct.

(4) If $\int_0^x f(t)\,dt=\frac{x^4}{2}$, then $\int_0^4 \frac{1}{\sqrt{x}} f(\sqrt{x})\,dx =$

(A) 8 (B) 16 (C) 128. (D) 256.

(5) The number of real roots in $(0,1)$ for $5x-2-\int_0^x \frac{dt}{1+t^8}=0$ is

(A) 0 (B) 1 (C) 2 (D) greater than 2.

2. Fill in the blanks.

(1) If $f(x)=(x^2+1)(x^2+2)(x^2+3)(x^2+4)$, then $f^{(6)}(0)=$ _____.

(2) The average value for $f(x)=\cos^4 x$ on $[0,\pi]$ is _____.

(3) Using Simpson's Rule with $n=4$ to estimate $\int_2^4 \frac{1}{x-1}\,dx$, the approximation is _____.

(4) If $\lim_{x\to\infty}\left(\frac{x+a}{x-a}\right)^x=8$, then $a=$ _____.

(5) $\lim_{n\to\infty}\frac{1}{n}\left(\sqrt{1+\cos\frac{\pi}{n}}+\sqrt{1+\cos\frac{2\pi}{n}}+\cdots+\sqrt{1+\cos\frac{n\pi}{n}}\right)=$ _____.

3. Find the area of the surface generated by revolving the curve $4y=x^2\ (1\le y\le 3)$ about the y-axis.

4. Solve the following first-order linear differential equation

$$xy'-y=2x\ln x,\quad x>0.$$

5. If the line $y=x$ is tangent to the curve $y=\log_a x$, find the value of a.

6. A isosceles triangle is to be inscribed in a circle of radius R. What is the largest perimeter possible for the isosceles triangle? Please provide the reason.

7. Find all values for p such that the improper integral $\int_0^\infty \frac{e^{-x}}{x^p}\,dx$ converges.

8. Evaluate the following limits.

(1) $\lim_{x\to 0}\frac{(1+x)^{\frac{1}{x}}-e}{x}$.

(2) $\lim_{x\to 0}\frac{3\sin x+x^2\cos\frac{1}{x}}{(1+\cos x)\ln(1+x)}$.

9. Evaluate the intergrals.

(1) $\int_{\frac{1}{e}}^{e} \frac{\ln^2 x}{x}\,dx$

(2) $\int_{1}^{\sqrt{2}} \frac{1}{x^3\sqrt{x^2-1}}\,dx.$

(3) $\int_{1}^{\infty} \frac{1}{x^6(x^5+4)}\,dx$

(4) $\int \frac{1}{(1+x+x^2)^2}\,dx.$